

Python Lab Examination

Instructions

- Write a separate Python function for every question.
- Call each function to demonstrate its output.
- Use real-life data and appropriate list, tuple, and dictionary operations.
- Avoid writing the entire solution as a single block of code.
- Create VE and put all answers
- Submit only Github link

Question 1: Student Marks Analysis

A college stores students' marks in a list. Write a function that accepts a list of marks and displays the highest mark, lowest mark, average mark, and number of students who passed (marks of 40 or above).

Question 2: Online Mobile Store

An online store maintains a list of mobile prices. Write a function to display all mobiles costing more than ₹20,000, calculate the total inventory value, and find the most expensive mobile price.

Question 3: Shopping Cart Management

A customer adds products to a shopping cart. Write a function that accepts a list of product names and performs the following operations:

- Add a new product.
- Remove a purchased or unwanted product.
- Display the final cart.
- Count the total number of products.

Question 4: Employee Salary Analysis

A company stores employee salaries in a list. Write a function to calculate the average salary, display salaries above the average, and increase every salary by 10%.

Question 5: Social Media Likes

A social media platform stores the number of likes received by five posts in a list. Write a function to find the post with the highest likes, calculate total likes, and display posts that received at least 100 likes.

Question 6: Hospital Patient Queue

A hospital maintains a list of patients waiting for consultation. Write a function that removes the first patient after consultation, adds a new patient to the queue, and displays the updated queue.

Question 7: Daily Expense Tracker

A person records daily expenses in a list. Write a function to calculate the total expenses, average daily expense, and the days on which expenses exceeded ₹1,000.

Section B: Python Tuple — Questions 8–13

Question 8: Student Registration Record

A college stores a student's roll number, name, course, and semester in a tuple. Write a function that displays each field separately and prints the complete student record.

Question 9: Mobile Product Details

An electronics store stores mobile details in a tuple containing brand, model, price, and storage capacity. Write a function to display the details and check whether the mobile price is below ₹30,000.

Question 10: GPS Location Tracking

A delivery application stores the latitude and longitude of a delivery location in a tuple. Write a function to display the coordinates and determine whether they represent a location in the approximate bounding box of Kolkata: latitude 22.4–22.8 and longitude 88.2–88.5.

Question 11: Product Order Details

An e-commerce website stores an order as a tuple containing order ID, product name, quantity, and unit price. Write a function to calculate the total order value and display the order details.

Question 12: Cricket Match Statistics

A cricket application stores a player's name, runs scored, balls faced, and number of boundaries in a tuple. Write a function to calculate the strike rate and display whether the player scored a half-century.

Strike rate = (runs scored / balls faced) × 100.

Question 13: Employee Attendance Record

A company stores an employee's ID, name, and attendance status in a tuple. Write a function that displays the record and reports whether the employee is present or absent.

Question 14: Student Result Management

A college stores student names and marks in a dictionary. Write a function that displays each student's result, calculates

Question 15: Social Media Profile Analysis

A social media application stores usernames and follower counts in a dictionary. Write a function to display all profiles, identify the user with the highest followers, and list users with more than 1,000 followers.

the class average, and identifies students who scored at least 75 marks.

Question 16: Mobile Store Inventory

A mobile store maintains a dictionary containing mobile names and their available quantities. Write a function that updates stock after a purchase, adds a new mobile model, and displays mobiles that are out of stock.

Assume a purchase must not exceed the available quantity.

Question 17: Online Shopping Bill

An online shopping cart stores product names and prices in a dictionary. Write a function to calculate the subtotal, apply a 10% discount when the subtotal is at least ₹5,000, and display the final bill.

Question 18: Employee Department Directory

A company maintains employee names and department names in a dictionary. Write a function to display employees department-wise and count the number of employees in each department.

Question 19: Product Sales Analysis

A business records the number of units sold for each product in a dictionary. Write a function to calculate total units sold, identify the best-selling product, and display products with sales below 10 units.

Question 20: Customer Support Complaint Tracker

A customer support team stores complaint IDs and their statuses in a dictionary. Write a function to:

- Add a new complaint.**
- Update the status of an existing complaint.**
- Count pending and resolved complaints.**

- **Display all complaint records.**
-

HINT

Question 1: Student Marks Analysis

A college stores students' marks in a list. Write a function that accepts a list of marks and displays the highest mark, lowest mark, average mark, and number of students who passed (marks of 40 or above).

```
def analyze_marks(marks):
```

```
    # Write your code here
```

```
    pass
```

```
analyze_marks([78, 35, 92, 45, 60, 28, 88])
```

Question 14: Student Result Management

A college stores student names and marks in a dictionary. Write a function that displays each student's result, calculates the class average, and identifies students who scored at least 75 marks.

```
def student_results(students):
```

```
    # Write your code here
```

```
    pass
```

```
student_results({  
  "Amit": 85,  
  "Priya": 92,  
  "Rahul": 67,  
  "Sneha": 78,  
  "Karan": 55  
})
```